Experiment 6: Deleting Files and Directories from HDFS (Windows)

Aim

To implement Hadoop file system shell commands on a Windows operating system to safely

remove individual files and clear directory structures within the HDFS cluster environment. Procedure
1. Open the Windows Command Prompt with administrative privileges to gain permission over

system configurations. 2. Run the status checking command jps to confirm that the active NameNode and DataNode cluster
storage daemons are up and running on your Windows machine. 3. List the current files and directories inside your targeted HDFS cluster using the hdfs dfs -ls
command utility. 4. Apply the basic deletion command hdfs dfs -rm followed by the absolute HDFS path to delete a
specific, individual target file. 5. Use the recursive parameter flag -r along with the deletion command to remove a targeted
directory along with all of its subfolders and files. 6. Implement the permanent deletion flag -skipTrash when running file removals to bypass the
default trash protection folder and free up cluster disk space instantly. 7. Re-run the directory listing command hdfs dfs -ls to confirm that the targeted items have been

successfully erased from the HDFS cluster storage layer. Commands

cmd
:: 1. Check your background storage daemon status on Windows
jps

:: 2. List items in your user storage path to find files for removal
hdfs dfs -ls /lab_workspace/student_records/data

:: 3. Delete a single text file from the designated HDFS path
hdfs dfs -rm /lab_workspace/student_records/data/backup_input.txt

:: 4. Attempt to clear a directory without a flag to observe the safety block hdfs dfs -rm /lab_workspace/student_records

:: 5. Delete an entire directory tree structure recursively using the -r flag

hdfs dfs -rm -r /lab_workspace/student_records

:: 6. Permanently drop an asset instantly and skip the trash safety folder
hdfs dfs -rm -skipTrash /sample_zone/old_data.txt

:: 7. Verify the final cluster storage structural configuration state
hdfs dfs -ls /

Use code with caution. Execution & Output

text
C:\hadoop\sbin>hdfs dfs -rm /lab_workspace/student_records/data/backup_input.txt Deleted /lab_workspace/student_records/data/backup_input.txt

C:\hadoop\sbin>hdfs dfs -rm /lab_workspace/student_records
rm: `/lab_workspace/student_records`: Is a directory

C:\hadoop\sbin>hdfs dfs -rm -r /lab_workspace/student_records
Deleted /lab_workspace/student_records

C:\hadoop\sbin>hdfs dfs -ls /lab_workspace/student_records
ls: `/lab_workspace/student_records`: No such file or directory